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RESEARCH ARTICLE

Effect of Stroop color word interference on selective attention task among healthy young adults

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ABSTRACT

Background: J Ridley Stroop showed that if someone was reading information and additional contradictory inputs were introduced, the person's reading pace would slow down. Stroop test is used to assess selective attention. Interference can occur between the controlled process and the automatic process. Psychologists have repeatedly established that the automatic nature of reading words, seeing that it is such a well-learned repeated activity can obstruct other tasks. There is additional evidence that practicing the Stroop task alters the extent of the interference effects. **Aim and Objectives:** The aim of this study was to assess the response time in Stroop test among healthy young adults and to compare the response time in Stroop test before and after the practice of word identification. **Materials and Methods:** Study included 82 2nd year medical undergraduates aged between 19 and 21 years, who gave their informed consent. Institutional ethical clearance was obtained. Stroop color-word test included three cards, the first bearing color patches, the second color words printed in congruent ink colors, and the third with color words printed in incongruent ink colors. The test required the subjects to name the colors on the first card, the words on the second card, and ink color on the third card without taking into account the printed words as fast as they can. Stopwatch was used to record the response time. Then, participants were asked to practice the above task of reading congruent and incongruent ink colors for about 6 days. After which they were again assessed for response time in Stroop test. Mean difference in response time between card two and three before and after the practice was tabulated and statistically analyzed. **Results:** The mean age was 20 ± 2 years. Before the practice session, the mean response time taken to read card three was significantly more (13.27 ± 2.66 min) compared to card two (6.26 ± 1.41 min) (P = 0.000, unpaired t-test). After the practice session, the mean response time taken was more for card three (12.14 ± 1.83 min) compared to card two (5.33 ± 1.21 min) (P = 0.000, unpaired t-test). However, there was no significant difference in response time before and after the practice session. **Conclusion:** The response time and interferences were more in incongruent Stroop test than congruent test. This indicates that the participants have delayed attention while performing Stroop test. This was in accordance with the Stroop effect. However, the study could not establish the effect of practice of word identification on Stroop test.

KEY WORDS: Practice; Response Time; Stroop Color Word Interference; Selective Attention Task

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INTRODUCTION

Attention is a system that allows us to choose and process important received information. Selective attention is the capacity to focus on one task at a time while filtering out any distracting external inputs. Divided attention, on the other hand, is the capacity to split one's attention between two or more tasks. Therefore, if one of the tasks is an automatic process, it becomes more simplistic to divide one's attention between the two tasks.[1] Stroop test is a highly reliable and reproducible test for the assessment of selective attention in cognitive psychology. In Stroop Color-Word test, subjects are required to name the color of the word printed in different ink colors, for example, (red printed in blue ink has to be read as blue). The reaction times (RTs) to name the ink color are slower when word's name is incongruent (word RED in blue ink) relative to when it is congruent with the color name (RED in red ink). Interference between word meaning and ink color which occurs prolongs response time. J Ridley Stroop (1900) demonstrated that if someone was reading information, and other conflicting inputs were added, the rate at which the person read would slow down which he called as Stroop Interference.[2,3] Interference can occur between the controlled process and the automatic process. Psychologists have repeatedly established that the automatic nature of reading words, seeing that it is such a well-learned repeated activity can obstruct other tasks. There is also an evidence that practice on the Stroop task itself changes the magnitude of interference effects. Stroop tasks involve the presentation of a stimulus that has conflicting sources of information; in a typical situation, the participant is required to name the color in which a word is printed when the word itself is a different color name called Stroop Effect.[3] There is also an evidence that practice on the Stroop task itself changes the magnitude of interference effects.[4] Despite the wide use of the Stroop in psychological research, there has been relatively little work on attention tasks in learning and education among young adults. With this background, the present work was carried out with objective to assess the effect of Stroop interference effect on selective task, that is, to assess the response time in Stroop test among healthy young adults and to compare the response time in Stroop test before and after the practice of word identification.

MATERIALS AND METHODS

This quasi-experimental study included 82 2nd year medical undergraduates aged between 19 and 21 years, who gave their informed consent. Institutional ethical clearance was obtained. Subjects included had normal eyesight or corrected eyesight with normal color vision. The subjects with chronic medical illness, neurological disorders, and color blindness were excluded from the study. The socio-demographic details were collected from the subjects before the application of Stroop color word test.

Stroop Color Word Test

Classic version of Stroop Color Word test.[2,3] was used. Stroop color-word test included three cards, the first bearing color patches, the second color words printed in congruent ink colors, and the third with color words printed in incongruent ink colors. The test required the subjects to name the colors on the first card, the words on the second card, and ink color on the third card without taking into account the printed words as fast as they can. Stopwatch was used to record the response time. Later, the participants were asked to practice the Stroop test task of reading congruent and incongruent ink colors for about 6 days. After which they were again assessed for response time in Stroop test. Mean difference in response time between card two and three before and after the practice was tabulated and statistically analyzed.

Data Analysis

The data were statistically analyzed using SPSS version 16. Mean and standard deviation was used to read the response time and student's *t*-test was used to assess the difference in response time between the card reading before and after the practice session. $P < 0.05$ was considered significant.

RESULTS

The mean age of the subjects in the study was 20 ± 1 year. Before the practice session, the mean response time taken to read card three was significantly more (13.27 ± 2.66 min) compared to card two (6.26 ± 1.41 min) ($P = 0.000$, unpaired *t*-test). After the practice session of 6 days, the mean response time taken was more for card three (12.14 ± 1.83 min) compared to card two (5.33 ± 1.21 min) ($P = 0.000$, unpaired *t*-test). There was no significant difference in response time before and after the practice session [Figure 1].

DISCUSSION

The present study findings showed that the participants took longer time to read incongruent Stroop card (card three) compared to congruent Stroop card (card two) which were statistically significant. Study by Khullar *et al*. on medical students also showed that the RT of incongruent trials was found to be significantly higher compared to the congruent trial in both groups ($P < 0.001$).[4] This is also in line with other previous studies done by Stroop *et al*.,[3] Ghimire *et al*.[5], and Macleod *et al*.[2]

An individual's performance on a basic task (e.g. reading names of colors) is compared with his or her performance on an analogous task in which a habitual response needs to be suppressed in support of an unusual one (i.e. naming the ink color that incongruously named color words are printed in). The increase in time taken to perform the latter task compared with the basic task is referred to as "the Stroop interference effect."[1] This interference effect from the word on naming its color demonstrates that we have problems in attending selectively to one component of a multidimensional stimulus and ignoring a highly over learned response to another.[6] Word reading is more practiced and hence more automatic

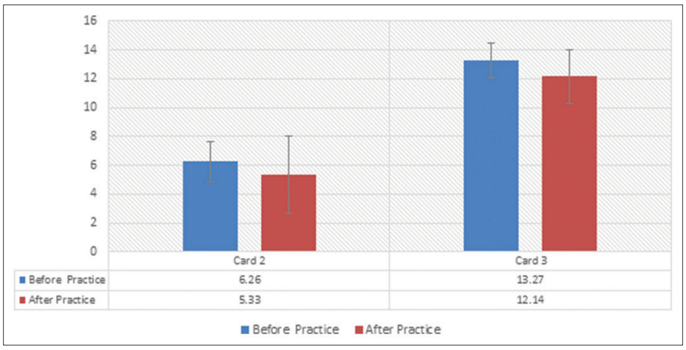

Figure 1: Distribution of study subjects (n=82) based on response time in Stroop test before and after the practice

and faster than color naming. Automatic word reading precedes and conflicts with color naming, hence leads to slower and erroneous responses to the color word stimulus.[1]

One hypothesis for Stroop interference is that of Speed of Processing Theory[3] which explains that the interference occurs because, people are able to read words much quicker and to name the color of the word is much more complex. The second theory is Selective Attention Theory where in interference occurs because naming the actual color of the words requires much more attention than reading the words.

Our study also showed that the participants took longer time to read incongruent Stroop card (card three) compared to congruent Stroop card (card two) before as well as after the practice session of 6 days. There was no significance of difference found between the both [Figure 1]. As per our observation in the present study, practice sessions in Stroop test among subjects were not monitored by the investigators and hence significant improvement might have not been occurred. The study by Gul *et al.* showed a situation where practice increases the size of congruency effect.[7]

The varied effects of interference on color and word naming imply that the amount of practice people have on tasks can modify Stroop effects. There is additional evidence that practicing the Stroop task alters the extent of the interference effects. In a study by Stroop after 8 days of practice, interference was reduced for conflicting words.[3] Similarly, Ellis *et al*. instructed their subjects to practice the Stroop color-naming task over three or four daily sessions. The results showed a significant decrease in interference over practice trials.[8] MacLeod also had participants practice on both word and color naming with Stroop stimuli across 5 consecutive days and reported that interference on color naming decreased by 49%.[9]

Possible explanations for the current findings include the observation that several of the participants seemed to deliberately slow down when reading color words, possibly to pay more attention. Others would read both lists at the same rate but make a mistake; realizing their error, and they would spend time rectifying it. These were the comments put forth by the study subjects. Studies on effect of practice on Stroop test among medical undergraduates are minimal; hence, we made an effort to assess for the selective attention using Stroop test. The limitations include less sample size, errors made while attempting the test were not counted, and practice session was not monitored in the study.

The Stroop Color word test is commonly used to assess cognitive interference ability; prior literature also cites its application to assess attention, processing speed, cognitive flexibility,[10] and working memory.[11] As a result, the Stroop Color word test might be used to assess multiple cognitive functions.

CONCLUSION

The response time and interferences were more in incongruent Stroop test than congruent test. This indicates that the participants have delayed attention while performing Stroop test. This was in accordance with the Stroop effect. However, the study could not establish the effect of practice of word identification on Stroop test. Further research is needed with larger sample size. The Stroop effect findings do have greater implications.

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